Gianmarco **Fabbri**

M.Sc. Cybersecurity Engineering Student · Cloud, Systems & Applied Cryptography

Torino, Italy

gianmarcofabbri2@gmail.com | Gianmarco-Fabbri | in gianmarco-fabbri

Summary

Second-year M.Sc. Cybersecurity Engineering student at Politecnico di Torino, with a B.Sc. in Computer Science and Engineering from the University of Bologna and a thesis on lattice-based post-quantum cryptography. Hands-on projects in Kubernetes runtime security and automated incident response, real-time systems and secure web development. Passionate about Cloud Security and the application of AI in cybersecurity.

Education

Politecnico di Torino — *Torino, Italy*

M.Sc. in Cybersecurity Engineering — *Oct. 2025 - Present*

- **Completed exams**: *Cryptography*, *Web Applications*, *Computer Architectures and Operating Systems*, *Hardware & Wireless Security*, *Optimization Methods and Algorithms*.
- **Study plan (in progress)**: *Networks & Cloud Technologies and Security*, *Information Systems Security*, *Security Verification and Testing*, *Threat Intelligence and Incident Management*, *Data Protection, Privacy and Anonymity*, *AI and Cybersecurity*, *Cybersecurity Laws and Regulations*, *Technology and Innovation Management in Cybersecurity*.

Alma Mater Studiorum – Università di Bologna — *Forlì-Cesena, Italy*

B.Sc. in Computer Science and Engineering — *Sept. 2022 - Oct. 2025*

- B.Sc. Thesis: *"The post-quantum era and lattice-based cryptography"* – Mathematical analysis of public-key primitives (RSA, ECC), quantum vulnerabilities (Shor's, Grover's), and post-quantum standards: **CRYSTALS-Dilithium** (digital signatures) and **CRYSTALS-Kyber** (key encapsulation).
- Rigorous engineering foundations in **Distributed Systems**, **Operating Systems**, **Database Systems**, and **Formal Methods**.

Experience

myDev s.r.l. (VEM Group) — *Forlì, Italy*

Software Engineer Intern — *Feb. 2025 - Mar. 2025*

- Engineered the **C#/.NET Core (ASP.NET)** backend for automated enterprise configuration management, applying the **Repository-Service Pattern** to enforce clean separation of concerns and robust data access layers.
- Developed responsive client interfaces in **Blazor (WebAssembly)** with **MudBlazor**, ensuring optimized client-side state handling, secure session workflows, and RESTful API integration.
- Conducted systematic integration testing, static analysis (SAST), and refactoring cycles to improve code quality and maintainability.

Projects

Compromised WordPress Site Recovery — *Professional firm (volunteer)*

Incident Response & WordPress Hardening – Linux Hosting, MySQL, Python — *2026*

- Investigated an SEO-spam compromise (**backdoors** in WordPress core, a malicious database payload, about **170,000** log events correlated in **Python**); rebuilt the site from official sources with **Docker** testing and hardened it with a **WAF**, **2FA** and credential rotation.

Kubernetes Automated Incident Response — *GitHub Repository*

Cloud Security, Runtime Monitoring & API Orchestration – Python, Kubernetes — *2026*

- Designed a Kubernetes security lab combining **Falco** runtime alerts, **Cilium** quarantine policies and workload replacement; implemented an authenticated Python handler with scoped RBAC and conflict-aware pod updates.
- Verified **16 regression tests** in a hardened Python 3.11 container; added CI build checks and cleanup controls that retain evidence until marked as exported.

PTL – Periodic Task Layer for FreeRTOS — *GitHub Repository*

Embedded Systems & Real-Time OS Scheduling – C, FreeRTOS, ARM Cortex-M3 — *2026*

- Co-developed a real-time middleware in **C** extending **FreeRTOS** on ARM Cortex-M3 (QEMU) with deadline-aware periodic task scheduling, overrun policies (SKIP, KILL, CATCH_UP), and automated regression tests.

Theater Booking System — *GitHub Repository*

Full-Stack Web Architecture, 2FA & Concurrency Control – React, Node.js, SQLite — *2026*

- Developed a secure full-stack web application in **React** and **Node.js/Express**, implementing Role-Based Access Control (**RBAC**), Two-Factor Authentication (**2FA/TOTP**), and strict reservation concurrency control.

Videogame Planet Shop — *GitHub Repository*

Relational Database Design, Normalization & Data Access Layer – MySQL, Python — *2024*

- Architected an e-commerce database normalized to **Boyce-Codd Normal Form (BCNF)** with triggers and referential integrity for **ACID** guarantees; developed a modular multi-role **Python/Tkinter** desktop client.

100 Days War - Game Engine — *GitHub Repository*

Object-Oriented Architecture - Java — *2024–2025*

- Co-engineered a decoupled game engine in **Java** within a 4-person team, applying **MVC**, unit testing, and design patterns (**Observer**, **Strategy**, **Command**, **Factory**, **Builder**) to ensure modularity and maintainability.

Skills

Cryptography & Trust	Public-Key Cryptography (RSA, ECC), Digital Signatures, Hash Functions, Lattice-Based PQC (Dilithium, Kyber), PKI, Verifiable Credentials Concepts
Cloud & Systems	Kubernetes, Docker, Linux Systems Architecture, eBPF/Falco, Cilium, CI/CD Pipelines, Microservices / RESTful APIs
Development	Python, C# (.NET Core / ASP.NET / Blazor WebAssembly), Java, C, SQL, PHP, JavaScript, Bash
Security & Auditing	Threat Modeling, API Security, SAST, OWASP Top 10, Network Protocols (TCP/IP), Wireshark, Burp Suite
Languages	Italian (Native), English (B2 – Cambridge B2 First), French (A2 – Basic)